Government of Karnataka









METRO PARKING POLICY

2025

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Definitions

a. **On-Street Parking-** available parking spaces for vehicles on a street along with the sidewalk, or anywhere on the street at available/ designated parking spaces for vehicles on a street.

b. **Off-Street Parking-** Off-street parking refers to available parking spaces for vehicles within an enclosed parking lot or garage. Parking lots may be owned by a Municipality, Municipal Corporation/Government organizations or privately owned.

c. **Parking Zones/areas -** a space where any kind of automobile can be parked.

d. **Parking Lot –** A designated place where one can leave their private vehicles/automobile for a specific period of time.

e. **Terminal Stations** - The first and last stations on a Metro corridor.

f. **Interchange Metro stations –** Interchange Metro Station is a Metro station that serves as a connecting point between two or more Metro lines or different modes of public transportation. It allows passengers to transfer from one line to another without exiting the station premises, making their journey smoother and more efficient

g. **Metro** – In this present context, Metro shall mean BMRCL premises/network.

h. **Mobility Management** - Mobility management can be broadly defined as creating and managing mobility options, at both the systemic and system-to-customer levels, to improve the reach, efficiency, and affordability of public transportation services.

i. **CBD** - **Central Business District (CBD)** is the **commercial and business hub of a city**, characterized by high-density development, tall buildings and a concentration of offices, retail outlets, financial institutions and services. It is often the most accessible part of the city, served by multiple public transportation modes.

j. **ITS** - **Intelligent Transportation System (ITS)** refers to the application of advanced **information and communication technologies (ICT)** to enhance the efficiency, safety, and sustainability of transportation systems. ITS improves traffic management, provides real-time information, and enhances the overall mobility experience for commuters and operators.

k. **First and Last mile connectivity**- The first and last stage of an individual trip – from their starting point to the nearest public transport station (first mile) and from the last station to

their final destination (last mile). Effective first and last mile solutions are crucial for ensuring seamless, convenient, and efficient public transportation.

l. **Micro-Mobility Vehicles**- Electric vehicles with a maximum speed of 25 kilometres per hour (kmph) and its equipped motor shall have a thirty-minute power less than 250 watt. These vehicles are approved by any authorized testing agency specified in the rule 126 of the Central Motor Vehicles Rules, 1989 as being compliant with the requirements specified in Notification no. G.S.R. 291(E) (2) dated 24.04.2014 of the Government of India.

m. **Transit Stations-** Bus terminals, Metro stations, Railway stations and large transportation hubs like Traffic Transit Management Centres, Multi-modal hubs that serve as a gateway to access public transport operations.

n. **Park and Ride- -** a designated parking area near public transportation hubs, such as Metro stations, bus terminals, or rail stations, where commuters can park their private vehicles (cars, bikes, or bicycles) and transfer to public transit for the remainder of their journey.

o. **Urban Gateway -** An Urban gateway is a key access point that connects a city to surrounding regions or the wider world through major transportation infrastructure like airports, railways, and highways. It facilitates the flow of people and goods, playing a crucial role in the City's economic and social connectivity.

BMRCL METRO PARKING POLICY FOR BENGALURU CITY

1. Background

Bengaluru is the capital city of Karnataka and is one of the fastest growing Cities in the Country. With the exponential increase in population of Bengaluru coupled with a rapid increase in the income level, there has been a consequent upsurge in number of registered vehicles plying in the City.

According to the data published by the Transport Department, Government of Karnataka, on an average 2500 number of vehicles are being registered in RTO each day, the number of registered vehicles in the city as on March 2024 has crossed 1 crore (10million) and the compounded annual growth rate of vehicle registration is over 10% per annum.

According to a report by TomTom, the Netherlands-based global provider of navigation, traffic and map products, Bengaluru ranks a low 149 (of 231), on the 2019 Mercer's Quality of Living Index, it is one of the world's most traffic-congested city. Average speed on critical roads drop to less than 10 km per hour during peak traffic hours and travel time are 48% longer than during the baseline non-congested conditions. This enormous trend of rise in numbers of vehicles is creating pressure on the parking needs of the City which is in turn affects the demand for parking at Metro stations as well.

Bangalore Metro, also known as Namma Metro, is a Mass Rapid Transit System (MRTS) that has transformed the way people commute in Bengaluru. Bengaluru, often referred to as the Silicon Valley of India, is known for its traffic congestion. The Metro Rail system has provided a much needed respite to the city's residents by reducing the number of vehicles on the roads and by reducing the pollution caused by the traffic congestion.

The total sanctioned network of BMRCL of Phase-1, Phase-2, Phase-2A, Phase-2B and Phase-3 is about 220.20 Km. Bangalore Metro Rail Network of Phase-1 and part of Phase-2 comprising a total length of 96.1 Km with 83 Metro stations is already under operation with an average daily ridership of about 9.5 lakhs commuters. The remaining portion of Phase-2 of 21.26 Km length is under construction. In addition, the Phase-2A Metro corridor from Central Silk Board (CSB) to K R Puram along the Outer Ring Road for a route length of 19.75 kms with 13 Elevated Stations and the Phase-2B Metro corridor from K R Puram to Kempegowda International Airport (KIA) for a route length of 38.44 kms with 16 Elevated +1 underground stations are under construction and Phase 3 Metro corridor of a route length 44.65 kms with 31 stations is under construction. With the increase in the patronage

of Metro commuters, the demand in parking at the Metro stations has also increased substantially. BMRCL has attempted to provide parking space at its Metro stations so that commuters can park their vehicles and take the Metro towards their destination. At present, BMRCL has a total area of 1.15 lakh sq.m area of parking space at 69 Metro stations (80 locations) with parking facility for bicycles, 2-wheelers, 3-wheelers and 4-wheeler vehicles.

2. Parking Policy 2.0 for Bengaluru City

The Government of Karnataka has approved Parking Policy 2.0 for Bengaluru and the Directorate of Urban Land Transport (DULT) has prepared the Area Parking Plans for all zones in Bengaluru (BBMP Area) in accordance with the said parking policy. The overarching principle of the Parking Policy 2.0 emphasises on reducing the demand for parking and facilitating organized on-street parking so that the road space can be efficiently utilised for transit and NMT supportive uses. The policy suggested paid parking to discourage the use of private mode and thereby ensuring modal shift to public transport, development of off-street parking facilities through market driven parking supply and management mechanism and enforcing strict regulations to manage the parking demand.

Further, the Parking Policy 2.0 has also recommended park-and-ride (off-street) near peripheral Mass Transport stations and recommends to encourage pricing at the stations in such a way that park-and-ride users are segregated from general parking users by allowing subsidized long duration fee for park-and ride users.

3. Need for Metro Parking policy

A growing trend in parking demand is seen in the entire city, especially near the transit stations. Haphazard on-street parking near the Metro stations not only creates chaos in the adjacent residential layouts but also reduces the capacity of the arterial roads with vehicles parked on either sides.

In addition to this, the lack of safe, affordable and reliable first and last mile connectivity options along with poor pedestrian infrastructure facilities discourages the Metro commuters in patronising Metro and as their preferred mode of transport, thereby decreasing the Public Transport patronage.

In view of these concerns and in the absence of a well-organized and well-planned parking regime around the Metro stations, a clear established Metro Parking Policy and framework to manage parking has become necessary.

Therefore, the Policy focuses to achieve the following objectives: -

1. Creation of Park and Ride facilities for the identified Metro stations.
2. Streamlining and enhancing the utilization of parking to ensure that the parking space provided by BMRCL is utilised by Metro users only.
3. To ensure safety and security of commuters and their vehicles at parking spaces.
4. To reserve parking for specific user groups such as cyclists and differently abled.
5. To determine the framework for Parking charges and Enforcement/Monitoring
6. Effective management of parking space through the use of Smart Technology

The parking policy will be reviewed periodically and the requirements will be amended overtime as and when first and last mile access to stations by walk, cycle, shared mobility or feeder buses are improved.

OBJECTIVE 1: PARK AND RIDE FACILITY FOR IDENTIFIED METRO STATIONS

The unprecedented growth in population and consequently in the number of vehicles is leading to the infamous traffic congestion in Bengaluru. Although Metro provides an efficient public transport alternative compared to private modes, private vehicle users have largely refrained from shifting to Metro as their preferred travel mode. A major reason for the same has been identified as the severe lack of reliable first and last mile connectivity to and from the Metro stations. Along with feeder bus service and good pedestrian infrastructure, the Park and ride system serves as an important aspect for first mile connectivity for specific station typologies to not only increase the Metro ridership but also to deter people from using private modes for long haul commute. The following parameters will be used to determine the potential for park and ride service at Metro stations:

1. Park and Ride at Peripheral/Terminal Stations/Urban Gateway

All **Peripheral/Terminal Metro Stations** have to be provided with Park and Ride facilities since these stations cater to large catchments from suburban areas who come to work in the city centre/CBD areas. Instead of driving all the way to their workplaces in the city centre/ CBD areas, commuters can conveniently park their vehicles at these stations and transfer to the Metro train/s. This approach not only alleviates traffic congestion in the

urban core but also promotes the use of Public transportation, leading to a more efficient and sustainable commute.

Implementation strategy

a. BMRCL will have to compulsorily acquire/purchase about 1 to 2 acres of additional vacant land/s at all terminal Metro stations for building of parking spaces/lots.

b. In case BMRCL is unable to acquire 1 to 2 acres of vacant land, BMRCL has to explore the possibility of providing multilevel basement parking for creating such parking facilities adjoining the Terminal Metro Stations.

c. **Approach State Authorities**: BMRCL will approach State Government to direct the relevant authorities (e.g., BDA, BBMP) who own lands near Metro stations to transfer ownership of land to BMRCL for the development of park and ride facilities.

d. **Collaboration with Private entities**: If adequate Government lands are not available, BMRCL will explore purchase or leasing of vacant Private lands. This will require a detailed analysis to ensure that the land is ideally located for park-and-ride use. This acquisition process may include direct negotiations with landowners or bidding, depending on the local regulations/permitted process.

e. **Central Government assistance**: BMRCL may request for financial assistance from Central Government in the form of VGF for land acquisition, in order to reduce the financial burden on BMRCL.

f. Explore innovative financing models, including **Land value capture** and **congestion pricing**, to help fund the development of park-and-ride facilities. Additionally, revenue from **advertising** within parking areas can offset operational costs, if viable and feasible.

g. **Grants and Subsidies**: BMRCL will request for **grants** or **subsidies** under programs aimed at reducing traffic congestion and promoting the use of public transport. For ex: the **FAME India Scheme** promotes the adoption of electric vehicles (EVs) and the development of EV charging infrastructure. It also includes support for eco-friendly transport initiatives, which could integrate with park-and-ride facilities. BMRCL may request subsidies to develop **EV charging/swapping stations** at few Metro stations for park-and-ride facilities, making them more attractive for users with electric vehicles. Similarly, the **National Smart Mobility Mission** supports projects that aim to reduce

congestion, promote eco-friendly modes of transport, and integrate multi-modal transport. BMRCL may apply for funding to integrate park-and-ride facilities into a Multimodal transport system.

2. Parking facilities at interchange Metro stations

Interchange Metro Stations, which serve as critical nodes within the Metro network where multiple Metro lines intersect, require well-planned and efficiently managed parking facilities. These stations experience high passenger volumes due to their role as transfer hubs for commuters traveling between various lines to reach their final destinations. Providing adequate parking facilities at interchange Metro stations serves multiple purposes like Enhancing Multimodal Integration, Facilitating Seamless Transfers, Reducing Traffic Congestion and Pollution, significantly boost Metro ridership.

Implementation Strategy

a. **Basement parking**: At interchange stations, BMRCL will explore provision of basement parking within the station area. To achieve this, stations will have to be planned and designed during the initial design stage itself to incorporate provisions for basement parking. This may involve additional cost of construction of such Stations, which may have to be suitably provided for in the DPR.

b. **Over the road parking:** BMRCL has constructed stations and viaducts with minimal road occupancy to prioritize train movement. Where basement parking is unfeasible, **over-the-road parking** can be explored after assessment of cost/benefit analysis and making sure that potential bottlenecks are not created in traffic flow, using ramps or lifts to optimize space and meet parking demand efficiently.

c. **Integrating with Multi-level parking in the vicinity of Metro station:** BMRCL will integrate with the nearby Multi-level parking (if available) and ensure the facilities are safe, well-lit, and equipped with smart parking solutions and plan for seamless connectivity between parking areas and Metro stations, with well-designed pedestrian pathways/travelators and signage. The roads in and around Metro stations will be kept free for improving approach to the stations. Suitable directional traffic on such roads will be planned and implemented.

a. **Approach State Authorities**: BMRCL will work with the State Government to direct agencies like BDA and BBMP to allocate Government-owned lands near interchange Metro stations for park-and-ride development.

b. **Collaboration with Private sector**: Where Government lands are not available, BMRCL will engage with private landowners for potential acquisitions or partnerships, possibly through public-private partnerships (PPP).

3. Park-and-Ride Facilities in Residential Areas with High Population Density and Narrow Roads

Metro stations surrounded by thickly populated residential catchment in the radius beyond 1 km and up to 3 kms with no direct Metro connectivity will be equipped with Park and Ride facility at the Metro station to enhance accessibility and encourage use of the Metro system. The 1 km radius to be determined through a walk/cycle-shed instead of a 'as crow flies' distance.

In addition, the areas with narrow residential roads pose significant operational challenges for standard 12-meter BMTC buses, hindering effective public transport access. In the absence of a regulated last-mile connectivity solution, commuters struggle to reach Metro stations. Therefore, it is essential to implement parking facilities at strategically located Metro stations, particularly those with a large catchment area, to enhance overall accessibility and connectivity. To supplement these efforts, e-Autos can be encouraged to provide last mile connectivity to commuters residing in these areas.

<u>Implementation Strategy</u>

a. **Private sector collaboration**: For Metro stations facing land constraints for parking, BMRCL will explore **Public-Private Partnerships (PPP)** with private developers, shopping malls, office complexes, and other commercial establishments located near the Metro stations (preferably within 100m). This could involve negotiating **shared-use agreements** where parking spaces at these locations are utilized by Metro commuters during weekdays and allocated to mall and other users during weekends or non-peak hours. This approach optimizes existing parking infrastructure, alleviates land acquisition costs, and enhances the convenience for both Metro users and private sector stakeholders.

b. **Incentivizing Higher FAR for Commercial Establishments:** BMRCL will request the State Government to provide incentives in the form of higher Floor Area Ratio (FAR) to the private entities in the vicinity of the Metro stations allowing them to develop additional built-up area beyond the usual zoning limits. This additional FAR can serve as an incentive for private entities to collaborate with BMRCL in developing or

sharing parking facilities for Metro commuters. The FAR used for parking infrastructure developed as part of the collaboration shall **not be counted towards the total FAR limit.** This allows the private entities to allocate sufficient space for parking facilities without compromising with their developmental potential.

4. Parking at Metro Stations near major multimodal transportation Hubs

Metro stations that are located within close vicinity to other major transportation hubs, such as Railway stations, bus stands, and airports, will explore the potential for multi-level parking facilities. Implementing such facilities can significantly enhance the interconnectivity between different modes of transportation, improve commuter convenience, and reduce congestion.

Implementation Strategy

Sharing parking space with other Transport authorities: BMRCL will collaborate with relevant authorities (e.g., Railways, BMTC, KSRTC, KIA) to acquire or make use of the underutilized Government-owned lands near Metro stations. If acquisition is not possible, BMRCL can jointly share the parking facilities with other agencies, allowing users of both the Metro and other modes of transport to park their vehicles and seamlessly access multimodal transportation service.

5. Metro Stations with Limited thorough access

Metro stations located near large tech parks, defence areas, lakes, airports, or other expensive land parcels often face significant challenges in terms of connectivity and accessibility. These areas typically lack **adequate thoroughfare access**, including **pedestrian-friendly pathways**, **dedicated cycling tracks**, or **reliable feeder services** such as shuttle buses or auto-rickshaw networks. As a result, commuters residing beyond walkable or cyclable distances often find it difficult to access Metro stations conveniently. Implementing a well-planned parking facility at such stations is essential to address first- and last-mile connectivity challenges for commuters in these areas.

Implementation Strategy

Integrate the parking facilities with the land owned by the Stakeholders: Collaborate with stakeholders like tech parks, defence establishments, and airport authorities to

identify access issues and develop joint parking spaces at the nearby locations or within their premises.

6. Collaboration with Private entities

For Metro stations facing land constraints for parking, BMRCL will collaborate with private entities such as shopping malls, office buildings, and hotels that are located in the vicinity of the Metro stations and have substantial parking facilities. BMRCL will enter into agreement with these private players to provide shared parking spaces for Metro commuters. This approach optimizes existing infrastructure and enhances commuter convenience. In fact, the National Urban Transport Policy (NUTP), 2006 at the Government of India level states that '**Multi-level parking complexes should be made a mandatory requirement in city centre(s) that have several high rise commercial complexes and these can come up through public-private partnerships**'.

Implementation Strategy

Private sector collaboration: For Metro stations facing land constraints for parking, BMRCL will explore **public-private partnerships (PPP)** with private developers, shopping malls, office complexes, and other commercial establishments located near Metro stations. This could involve negotiating **shared-use agreements** where parking spaces at these locations are utilized by Metro commuters during weekdays and allocated for malls and other users during weekends or non-peak hours. This approach optimizes existing parking infrastructure, alleviates land acquisition costs, and enhances the convenience for both Metro users and private sector stakeholders.

OBJECTIVE 2: STREAMLINING PARKING TO ENSURE THAT THE PARKING SPACE PROVIDED BY BMRCL IS UTILISED BY METRO USERS ONLY

The Metro parking space is sometimes utilised by the people who are not the actual Metro users/commuters but might be owning a commercial/retail outlet nearby the Metro station. In such cases, the non-Metro users park their vehicle/s in the Metro parking lot and the Metro commuter is forced to park his/her vehicle on road due to lack of parking space. However, if non-Metro users are prohibited from parking, their use of private vehicle would be

disincentivised due to difficulty in finding parking space, thereby, automatically incentivising the use of Metro to access these areas.

Therefore, it is important to devise provisions/ regulations to ensure that Metro parking space is solely utilised for Park and Ride by the Metro users/commuters only, for which:

1. **Automated Access Control Systems:** Mechanisms will be developed to segregate park-and-ride users from general parking users at these locations. This can be also done by integrating the Metro travel card with the parking charges. Such integration will also be facilitated through other digital ticketing means like QR code, FASTAG etc.
2. The charges will be deducted accordingly at the time of exit of the vehicle – normal charges are deducted if Metro services are used between parking in and parking out, and penalty charges if the card is not used for Metro travel excluding the retail shop owners within the Metro premises. The normal and penal charges will be as decided by BMRCL from time to time which will have to be duly notified/displayed prominently.
3. This infrastructure will also ensure
    a. Well managed parking
    b. Direct relay of data to BMRCL
    c. No leakages in fare collection
4. In addition, on-street public parking to be strictly prohibited within 250 meters of the Mass transit stations in accordance with the approved Bangalore Parking Policy 2.0.

OBJECTIVE 3: TO ENSURE SAFETY AND SECURITY OF COMMUTERS AND THEIR VEHICLES AT PARKING SPACES

Parking spaces often tend to be considered unsafe by women due to lack of activity, and eyes especially after dusk/late evening times. Therefore, while provisioning for parking, the following needs to be mandated to ensure security for both the users and their vehicles parked:

1. All parking lots need to be under CCTV surveillance 24x7 with adequate lighting installed at strategic points (**Signage- Annexure 1, Figure 4**). A storage capacity of up to 30 days of recording must be maintained to carry out the required audits.
2. The CCTV recordings will be integrated to the existing CCTV coverage of station area and will be accessible to Metro officials only so that any abusive incident by the parking contractors towards the commuters can be monitored. In addition, provision will be

made to stream the CCTV footage at the BMRCL central Control Centre for the purpose of monitoring.

3. Fire incident: To handle the fire related incidents at parking lots, the following facilities /equipment will have to be ensured:
    - Provision of sufficient fire buckets on a metal stand filled with fine and dry sand with water drums near the parking area
    - Provision of fire extinguishers at locations as per stipulated design standards.
4. **The emergency point of contact**: The parking lots will be managed by the operators of the nominated agency/contractor including the matters related to the damage to or theft etc. of vehicles parked and resolution of any other disputes with the customers etc. However, in case of emergency like fire, flooding etc. affecting the safety & security of Metro property or commuters, the Station Controller along with station security staff will be called for help by Parking lot operator and also for the further coordination with civic authorities etc. as the case may be. (**Annexure 1, Figure 5**).
5. In addition, the parking lots needs to be well paved and sheltered to the extent possible in order to avoid any kind of damage to the parked vehicles from wear and tear/sunlight/rainfall etc.
6. It is also essential that the parking lot includes clear markings and designated spaces for cycles, two-wheelers, and four-wheelers, with prominently visible entry and exit signs. These markings should ensure smooth ingress and egress without conflicting with other vehicles. Ingress and egress to the parking lots should ideally be avoided from major or arterial roads. Wherever possible, access should be provided via roads with lighter traffic, such as cross streets or service roads, to prevent vehicles entering or exiting the parking lots from disrupting the overall traffic flow in the area.
7. **Lighting and Visibility Enhancements:** Parking lots must be well illuminated to avoid any kind of illegal/criminal activities at night. In addition to this, the enhanced visibility makes the commuters especially the female commuters feel safe while dropping off/picking up their vehicle.
8. Professionally trained and experienced personnel to be deployed for manning the parking site and presence of a security staff 24X7 to ensure safety and security of vehicles.
9. Behaviour of the parking contractors to be monitored through CCTV by BMRCL parking in-charge officer and the contractor/s may be replaced immediately if found violating the code of conduct.

10. The parking contractors shall not charge excess amount than determined by BMRCL. In case of overcharging suitable penalty will be levied in addition to banning such operators
11. The Parking Contractors/Agency shall obtain adequate Third Party Liability Insurance for the stipulated Sum Insured Limit and terms of BMRCL, to cover for any risk of damage/theft of the vehicles parked in the Parking Lot/s.

OBJECTIVE 4: TO RESERVE PARKING FOR SPECIFIC USER GROUPS

1. Parking for cycles – A minimum of 10 parking stands for cycle parking will be provided in each Metro station parking lots at free of cost.
2. Parking for differently-abled
    - A minimum of one space for both two-wheeler and three wheeler parking for differently-abled will be earmarked in each Metro station parking lot with signage. In parking lots where car parking for more than 20 cars/four-wheelers is provided, at least 1 car/four-wheeler parking space shall be reserved for differently-abled. The identified space shall be near the entry gate of the parking lot and / or any other convenient location within the parking lot from where ramp for accessing station entrance / lift is the nearest.
    - The space will be provided as per Bureau of Indian Standards, National Building Code of India of 2016, IRC: SP:12:2015 and Harmonized Guidelines and Space Standards for barrier free built environment for persons with disability and elderly persons, 2016 of Ministry of Urban Development, Government of India.
    - The parking spaces will be indicated by using paint as required in above-mentioned guidelines. The signage to be used for different modes of transport are shown in **Annexure 1, Figure 1.**
3. Parking for BMRCL employees and official vehicles for maintenance & operation activities - Separate parking space will be reserved for vehicles of BMRCL employees/BMRCL Official vehicles at every station other than the commuter parking. No charges shall be levied for such vehicles of employees / O&M Vehicles.
4. Space for pick-up and drop-off for Micro mobility modes/other modes will be located on cross- streets or along the service roads near the Metro station with legible signage **(Annexure 1, Figure 6).**

5. Pre-fixed Auto Rickshaw booths- Auto rickshaws plays a crucial role in last mile connectivity and serve short trips in the city of Bengaluru. BMRCL has carved out dedicated parking space for Pre-Fixed Auto rickshaw booths at few Metro stations and has planned to expand it to all the Metro stations wherein the commuters can hire autos from the transit station to their destination by Pre-fixing their travel cost based on the distance travelled.

OBJECTIVE 5: PARKING CHARGES AND ENFORCEMENT/MONITORING

1. The BMRCL Fare chart will be displayed at the entry of the Parking lot to ensure that the parking Contractors/Agencies do not charge extra for parking. The signage indicating the dimensions of the fare chart board will be as shown in **Annexure 1, Figure 2**.
2. Park-and-ride users will be charged subsidized long duration fee for parking their vehicles at these facilities that nudge commuters to take transit than driving their own vehicles in the City.
3. Exploring possibility of increasing parking capacity by adoption of multi-level parking and automated parking systems will be considered.
4. Revision of the base parking fee to be done whenever a need is felt based on demand, availability of land etc.,
5. Abandonment of vehicles beyond the operational hours will be monitored through CCTV cameras. If a vehicle is parked for more than 3 days, the licensee /organisation (whoever is responsible for managing the parking space) must take adequate action to remove the vehicle from the parking lot by issuing notice to the vehicle owner by obtaining details of the ownership of the vehicle. In case of no response, a final notice will be issued stating that the vehicle will be auctioned.
6. Parking of vehicles after closure of revenue services: If a commuter of Metro railway desires to use parking facility after revenue closure, (S)he shall share his identity such as address, phone number, vehicle registration details/license or any such document or any other document as specified by the Metro authority—to the parking in-charge, along with the intended duration of parking. If a vehicle remains unclaimed for more than three days, it will be removed and auctioned as per Motor Vehicle Act. Additionally, vehicles not retrieved before the end of daily revenue operations will incur extra charges.

7. The redressal cell of BMRCL will address the complaints related to parking on regular basis. The complainant should be able to submit his/her grievances through online mode also.
8. Any damage to the vehicles due to the natural calamities and subsequent invocation of Insurance cover will be managed by the licensee.
9. Cleanliness: The licensee shall maintain the premises clean and tidy at all times. He shall arrange to mop the premises periodically.

OBJECTIVE 6: EFFECTIVE MANAGEMENT OF PARKING SPACE THROUGH USE OF SMART TECHNOLOGY

Increasing number of vehicles at the Metro station parking lots along with the mismanagement of available parking spaces lead to the parking related problems.

For efficient management and enforcement of parking, it is crucial to adopt technology for modernizing parking operations and to reduce monetary pilferage: Few of the technological solutions which may be considered by Metro based on patronage of parking lot and economical considerations are:

1. **Mobile application**: The availability of the parking slots can be displayed in the mobile app as well in a display near the Parking lot. A loop detector system needs to be installed in the parking lot to count the number of parking slots and indicate the slots available for parking. The information like availability of parking space at particular Metro station, will be displayed near the Parking lot as well as integrated with the Namma Metro app wherein the users can search for the available parking slot at their selected Metro station and based on the availability they can choose to take out their vehicle. A robust, technology-driven parking fee collection system will be implemented along with a loop detector system to eliminate pilferage and ensure accurate data recording. The system will capture details such as the number of vehicles parked, empty slots available for parking, parking duration, vehicle types, and timestamps, providing comprehensive records for auditing and facilitating data-driven planning to improve parking services.
2. **LED display representing the vacant slots:** Every Metro station will be provided with graphical interface to display the real time parking availability map. The map will indicate vacant parking slots (cycles, two wheelers and four wheelers separately) in green colour and occupied parking slots in red colour at a particular Metro station and also one

station before and after it so that the commuter can choose where to park his/her vehicle if his selected station does not have space for parking.

3. **Boom barrier:** To avoid the revenue loss from the parking contractors, the technology of boom barriers may be utilised as per the space available, wherein the device will be installed at the entry/exit of the Parking lot and it reads the processed information regarding the timing and number of the vehicle, time till which the parking is done and delivers a receipts which then can be paid by the commuter either through Namma Metro app (after integrating) or any other payment gateway like GooglePay/PhonePe/Paytm/FASTAG etc.

   Further, the National Urban Transport Policy (NUTP), 2006 at the Government of India level states that '**All parking complexes will be encouraged to go in for electronic metering so that there is better realization of parking fees to make the investments viable and also a better recovery of the cost of using valuable urban space in the parking of personal motor vehicles**'.

4. **Central Server:** A central server system is the heart of the smart parking management and guidance system. All the relevant data from the cameras, sensor, parking meter and mobile application will be collected, processed and transmitted from the central server to the display systems and parking meters. All the parking related information of all the stations can be integrated into central parking portal which will be managed and monitored from one central location by the BMRCL in-charge officer.

   In addition, BMRCL will integrate its parking data with the citywide parking portal to enable seamless and centralized parking management. This integration will ensure a cohesive approach to parking operations across the City.

5. **Start-ups/operators** will be encouraged to offer competitive and creative technology solutions that would help BMRCL to optimise the use of parking spaces. Engaging in **Public-private partnerships** can be a strategic approach for developing and managing parking facilities. Such collaborations will bring in investment, expertise, and innovation, leading to better-managed and maintained parking space.

ANNEXURE 1

Parking Signage

An effective signage System keeps the commuters informed of parking regulations. The following set of signage shows the standard to be followed while installing the sign boards that ensures legibility and uniformity:

1. **Signage's for parking for different modes**



Parking



Bicycle Parking



Two wheeler Parking



Car Parking



Parking for Differently abled

Figure 1: Signages for parking for different modes

2. **Signage for Parking fare chart board along with its dimensions**

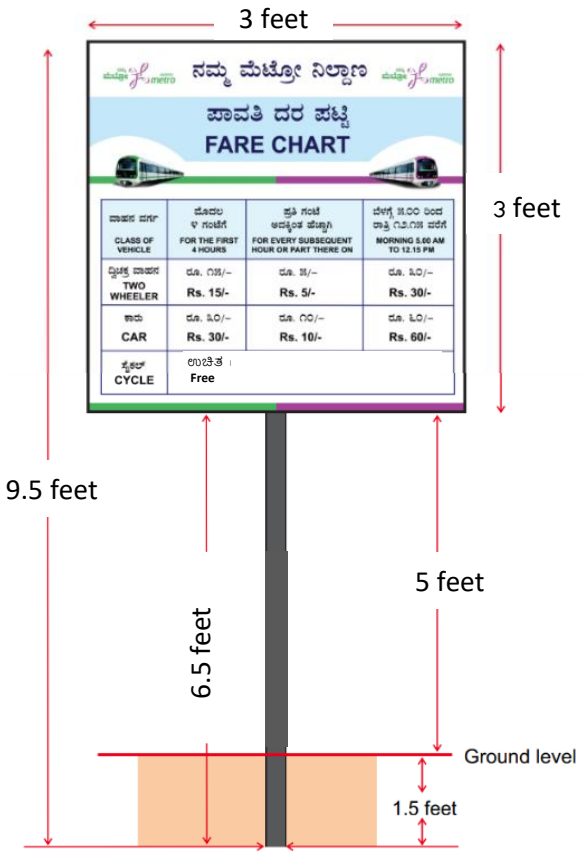


Figure 2: Signage indicating Parking fee along with the dimensions

3. **No parking board**: The signage board indicating that one cannot leave their vehicles unattended, any dropping off or picking up of passengers should be completed within 2 minutes and driven on as soon as possible. In case of parking in this zone, Automatic fines shall be levied with photographic evidence to the vehicle owner.



Figure 3: No Parking Board

4. **Surveillance signage:** This signage indicates that the corridor is under surveillance for monitoring the parking norm violations and security purposes. This board can be installed on the surveillance camera pole itself.



Figure 4: Surveillance Signage

5. **Helpline number**: Helpline number to be displayed at the Parking lot for any emergency situation.



Figure 5: Helpline signage

6. **Pick-up/drop off signage**: This sign shall indicate that the location at which this sign is placed is reserved for pick-up and drop-off passengers arriving by Car/Auto/Cab/Two-wheeler. Clock with number in centre icon provides information on the time for which vehicle is allowed to stand in that place for pick-up and drop-off passengers.

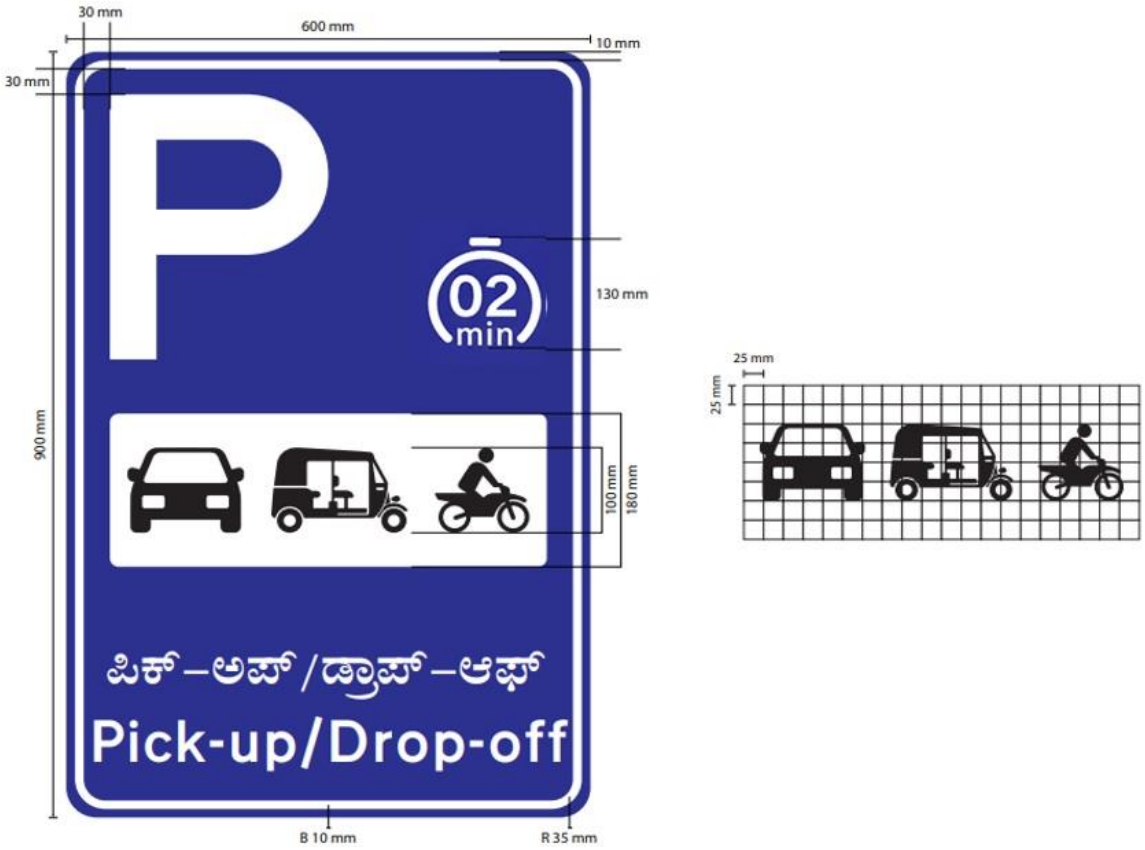


Figure 6: Signage for Pick-up/Drop-off